

Fig. 7: Select device options

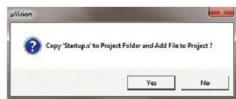


Fig. 8: Message window

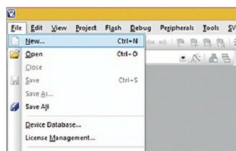


Fig. 9: Create New File window

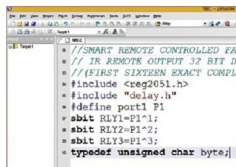


Fig. 10: NEC.c compilation file

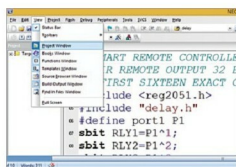


Fig. 11: Project window

fed to IC1 to maintain constant +5V DC at output. Constant output is fed to AT89C2051 MCU.

Construction and testing

An actual-size, single-side PCB pattern of the circuit using AT89C2051 MCU is shown in Fig. 3 and its component layout in Fig. 4.

Whenever you press
a button of the remote,
TSOP1738 (IRX1)

receives IR signals automatically and feeds it to Port3 pin P3.0 of AT89C2051. The controller receives the sequence of IR pulses and finds out when a button is pressed with the help of Assembly language program in the MCU. In this project we are using only four IR remote buttons, namely, Power, 1, 2 and 3.

Case 1. When you press Power button on the remote, the MCU sends logic low signal to T1, T2 and T3, and automatically making them cut-off (off). Whenever T1, T2 and T3 are in off state, relays RL1, RL2 and RL3 will also be in off state, and therefore the relays' contacts across the tapped resistors are open. Hence, there will be no voltage across the ceiling fan or bulb. This state is known as fan/bulb off state.

Case 2. When you press 1, immediately the MCU sends logic high signal to T1. This makes T1

to conduct and, in turn, energise relay RL1. As shown in Fig. 1, load (fan/bulb) is connected to 230V AC mains supply through tapped resistors R6 through R9. This should be taken as position 1 or low speed.

Case 3. When you press 2, immediately the MCU sends logic high signal to T2. This makes T2 to conduct and, in turn, energise relay RL2. Load (fan/bulb) is connected to 230V AC mains supply through tapped resistors R8 and R9. This should be considered as position 2 or medium speed.

Case 4. When you press 3, immediately the MCU sends logic high signal to T3. This makes T3 to conduct and, in turn, energise relay RL3. Load (fan/bulb) is connected to 230V AC mains supply without any resistor. This should be position 3 or full speed.

Case 5. When you press a button other than power, 1, 2 or 3, MCU will not take any action. It will simply insert a few delays (in milliseconds).

PARTS LIST

Semiconductors:

IC1	- 7805 voltage regulator
IC2	- AT89C2051 MCU
D1-D5	- 1N4007
T1-T3	- BC547 npn transistor
LED1	- 5mm LED
IRX1	- TSOP1738 receiver module

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

R1	- 220 ohm
R2	- 8.2-kilo-ohm
R3-R5	- 10-kilo-ohm
R6-R9	- 100-ohm, 2W
RNW1	- 10-kilo-ohm resistance array/network

Capacitors:

C1 - 2200 μ F, 25V electrolytic
C2 - 100 μ F, 16V electrolytic
C3 - 10 μ F, 16V electrolytic
C4, C5 - 33pF ceramic disk

Miscellaneous:

X1	- 230V to 12V centre-tapped transformer
CON1	- 3-pin berg strip male connector for X1
CON2	- 2-pin connector
CON3	- 2-pin connector
RL1-RL3	- 12V, 1C/O relay
X _{TAL}	- 11.0592MHz crystal
IR remote	- VIRE IR remote

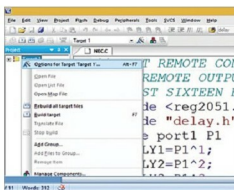


Fig. 12: Device Options window